EXPERIMENT-4

10/8/22

AIM: Setup and configure a LAN (Local Area Network) using a switch and ethernet cables in your lab.

How to set up a LAN :

Step 1 : Plan and Design an appropriate network topology taking into account network requirements and equipment.

Step 2 : You can take 4 computers, a switch with 8, 16 or 24 ports which is sufficient for networks of these sizes and 4 Ethernet cables.

Step 3 : Connect your computers to network switch via an Ethernet cable, which is as simple as plugging one end of the Ethernet cable into your co.

Step 4 : Assign IP address to your PCs.

1. Log on to the client computer as Administrator or as Owner.
2. Click Network and Internet connection
3. Right Click Local Area connection/Ethernet → Go to properties → Select Internet Protocol (TCP/IPv4) → Click on Properties → Select use the following ip address option and assign ipaddress.

IP Address Configuration Table

Device	IP Address	Subnet Mask	Connection
PC0	10.1.1.1	255.0.0.0	Switch Port 0
PC1	10.1.1.2	255.0.0.0	Switch Port 1
PC2	10.1.1.3	255.0.0.0	Switch Port 2
PC3	10.1.1.4	255.0.0.0	Switch Port 3
Switch	10.1.1.5	255.0.0.0	VLAN 1 Interface

Step 5: Configure the Switch Management IP

a). Click the Switch → CLI Tab.

b). Enter the following commands to assign a management IP

```
Enable
Configure terminal
Interface vlan 1
Ip address 10.1.1.5 255.0.0.0
No shutdown
Exit
```

Step 6: Verify connectivity using ping

a) Click a PC → Desktop Tab → Command Prompt

b). Use the ping command to test connectivity to the other PCs.

Eg: ping 10.1.1.2

c). Successful "reply from..." messages confirm connectivity.

Step 7: Demonstrate Data Sharing

a). Add a server device (End devices → Server PT) and connect it to switch.

b). Assign it an IP address. Desktop → IP Configuration tab

c) Open the server's services (config tab) and enable the HTTP service.

d). From another PC, open the Web Browser (Desktop tab) and enter the server's IP address to load the page, confirming successful data transfer across the LAN.

Step 8: Observe Packet flow in Simulation Mode

a). Switch to Simulation mode

b). Send a ping or web request between two devices and use the play/capture controls to step through the packet as it travels hop by hop through the switch, observing MAC-address based forwarding.

RESULT: Thus the setting up a Local Area Network (LAN) has been executed successfully.